



Three Tools to Help You Protect Your Network

The network administrator's constant fear is a compromised network due to hackers or unauthorised access. This article describes the use of Wireshark, NetworkMiner and Snort, three popular open source packet analysis tools that help to analyse hacking or cracking attempts on a network.

The task of monitoring and administering networks has become both important and difficult due to the huge amount of information flowing through different transmission channels. In any organisation, it's a challenging task for network administrators to analyse the financial, military, educational or social information passing through their networks. Network crackers are very keen to access the confidential data running inside a target's network. Hence, there is the need for very effective tools that can analyse hacking or cracking attempts. Generally, crackers analyse the opponents' networks and capture the information in their records. This task is classically known as network sniffing, by which the information flowing through a network is repeatedly analysed.

There are a number of software products available in the technology market that provide network sniffer modules, using which, the systems administrator can analyse the packets. Packet capturing is the procedure of capturing and logging movement. The packet analyser is also referred to

as a network analyser, protocol analysis tool or protocol analyser, packet sniffer, Ethernet sniffer or, simply, a wireless sniffer. Such software is technically a program that intercepts, seizes and logs the traffic passing through a network infrastructure. As information streams over the system, the sniffer catches every packet and, if required, translates the packet's crude information, demonstrating the qualities of different fields in the parcel.

Active and passive sniffing

Sniffing is a technique for fetching network information by capturing network packets. There are two types of packet sniffing in networks—active sniffing and passive sniffing. In active sniffing, the packet sniffing tool or software sends the requests over the network and then, in response, calculates the packets passing through the network. Passive sniffing does not rely on sending requests. This technique scans the network traffic without being detected on the network. It can be useful in places where networks are running critical